

Study Report: Analyzing Data with Power BI and Power Pivot - Power BI Analyst

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Family: Power BI Analyst

Level: Advanced

Chapters: 7

Source: Reference: Analyzing Data with Power BI and Power Pivot - Marco Russo & Alberto Ferrari

Official sources and free libraries

- <https://www.sqlbi.com/books/analyzing-data-with-power-bi-and-power-pivot/>

Summary

Study report for *Analyzing Data with Power BI and Power Pivot* by Marco Russo & Alberto Ferrari (Advanced level) mapped to the Power BI Analyst role. Data modeling and DAX for complex enterprise models.

Chapters

Track: Book-Intro

Chapter 1: About Analyzing Data with Power BI and Power Pivot [Advanced]

Chapter focus: About Analyzing Data with Power BI and Power Pivot** *(Level: Advanced)

This study report summarizes *Analyzing Data with Power BI and Power Pivot* by Marco Russo & Alberto Ferrari for the Power BI Analyst role. The resource is rated Advanced level. Data modeling and DAX for complex enterprise models. Use this report alongside the official material to prepare for CodeQuest review.

Code Reference:

```
# Analyzing Data with Power BI and Power Pivot
# Author: Marco Russo & Alberto Ferrari
# Role: Power BI Analyst
# Level: Advanced
```

What it shows: Metadata block records the source book and target role family.

Topic guide

What it actually is

A structured study report based on Analyzing Data with Power BI and Power Pivot.

When to use it

When learning Power BI Analyst skills at Advanced level.

Where to use it

Power BI Analyst training paths and certification prep.

Real use example

A learner reads this report before starting the full book or free online guide.

Track: Book-Context

Chapter 2: Why This Book Matters for Your Role [Advanced]

Chapter focus: Why This Book Matters for Your Role** *(Level: Advanced)

Power BI Analyst professionals use ideas from Analyzing Data with Power BI and Power Pivot to solve real workplace problems. Data modeling and DAX for complex enterprise models. This chapter explains how the book fits into your learning path and what you should be able to do after studying it.

Code Reference:

Role: Power BI Analyst

Book focus: Data modeling and DAX for complex enterprise models.

Recommended level: Advanced

What it shows: Connects the book topic to the job role outcomes.

Topic guide

What it actually is

Role-aligned learning connects theory to job tasks.

When to use it

During career planning and syllabus design.

Where to use it

Power BI Analyst bootcamps and CodeQuest teacher assignments.

Real use example

A teacher assigns this book report before a advanced skills checkpoint.

Track: Book-Topics

Chapter 3: Key Topics Covered [Advanced]

Chapter focus: Key Topics Covered** *(Level: Advanced)

The main topics in Analyzing Data with Power BI and Power Pivot include practical concepts described as: Data modeling and DAX for complex enterprise models. Study each topic with hands-on practice. Take notes on definitions, workflows, and examples that match tools used in Power BI Analyst jobs today.

Code Reference:

- Topic 1: core concept
- Topic 2: core concept
- Topic 3: core concept
- Topic 4: core concept

What it shows: Topic list guides what to study chapter-by-chapter in the source.

Topic guide

What it actually is

Topic maps turn a book into actionable learning objectives.

When to use it

Before reading and while building chapter summaries.

Where to use it

Self-study, flipped classroom, and revision.

Real use example

A student maps each book chapter to a CodeQuest report section.

Track: Book-Applied

Chapter 4: Applied: Composite Models and Aggregations [Advanced]

Chapter focus: Composite Models and Aggregations** *(Level: Advanced)

Composite models mix Import and DirectQuery tables. Aggregation tables pre-summarize facts for drill-down performance. User hierarchies hit agg first, then detail - transparent to report users.

Code Reference:

```
Sales Agg = SUMMARIZE ( Sales, Date[Month], Product[Category], "Amount", SUM (
Sales[Amount] ) )
```

What it shows: SUMMARIZE builds an aggregation table DAX engine can substitute automatically.

Topic guide

What it actually is

Composite + aggregation optimizes large models without losing detail.

When to use it

When Import models exceed memory or DirectQuery is too slow for drill-down.

Where to use it

Global retail chains and multi-year telemetry models.

Real use example

Retail BI team uses agg table for monthly scans - detail loads only on drill-through.

Chapter 5: Applied: Advanced DAX: Context Transition [Advanced]

Chapter focus: Advanced DAX: Context Transition** *(Level: Advanced)

Calculated columns evaluate row-by-row; measures use filter context. Context transition occurs when CALCULATE iterates - a common source of bugs. Use variables (VAR) for readability and single evaluation.

Code Reference:

```
Rank in Category =  
VAR CatSales = CALCULATE ( [Total Sales], ALLEXCEPT ( Product, Product[Category] ) )  
RETURN RANKX ( ALL ( Product[Product] ), CatSales, , DESC, Dense )
```

What it shows: VAR stores intermediate results - easier to debug than nested CALCULATE.

Topic guide

What it actually is

Advanced DAX handles ranking, moving averages, and dynamic segmentation.

When to use it

When CALCULATE-heavy measures misbehave or rankings are needed in visuals.

Where to use it

Subscription analytics with tiered discount logic.

Real use example

VAR-based rank measure finally matches Excel prototype after context transition fix.

Chapter 6: Applied: Fabric Data Factory Pipelines [Advanced]

Chapter focus: Fabric Data Factory Pipelines** *(Level: Advanced)

Fabric Data Factory (successor patterns to ADF in Fabric) orchestrates copy, notebook, and Dataflow Gen2 activities on OneLake. Schedule pipelines for bronze ingestion before semantic refresh.

Code Reference:

```
# Pipeline: Copy blob -> Notebook cleanse -> Refresh dataset  
# Trigger: daily 05:00 UTC
```

What it shows: Orchestration ensures silver tables update before Power BI refresh fires.

Topic guide

What it actually is

Data Factory pipelines automate lakehouse ingestion in Fabric.

When to use it

When bronze/silver loads must run before semantic model refresh on OneLake.

Where to use it

Nightly ERP extracts and multi-source consolidation.

Real use example

Pipeline failure blocks dataset refresh - ops gets alert before executives see stale KPIs.

Track: Book-Plan

Chapter 7: Study Plan and Practice [Advanced]

Chapter focus: Study Plan and Practice** *(Level: Advanced)

Finish Analyzing Data with Power BI and Power Pivot with a weekly plan: read one section, write a summary, complete one exercise, and reflect on how it applies to Power BI Analyst. If a free edition exists, practice every example. Submit your notes and one mini-project demo for teacher review.

Code Reference:

```
Week 1: Read + notes  
Week 2: Exercises  
Week 3: Mini project  
Week 4: Review + quiz
```

What it shows: A 4-week plan turns reading into demonstrable skill.

Topic guide

What it actually is

Structured study plans improve retention and portfolio outcomes.

When to use it

After finishing the key topics chapters.

Where to use it

CodeQuest end-of-unit assessments.

Real use example

A learner completes the study plan and uploads a advanced project screenshot.